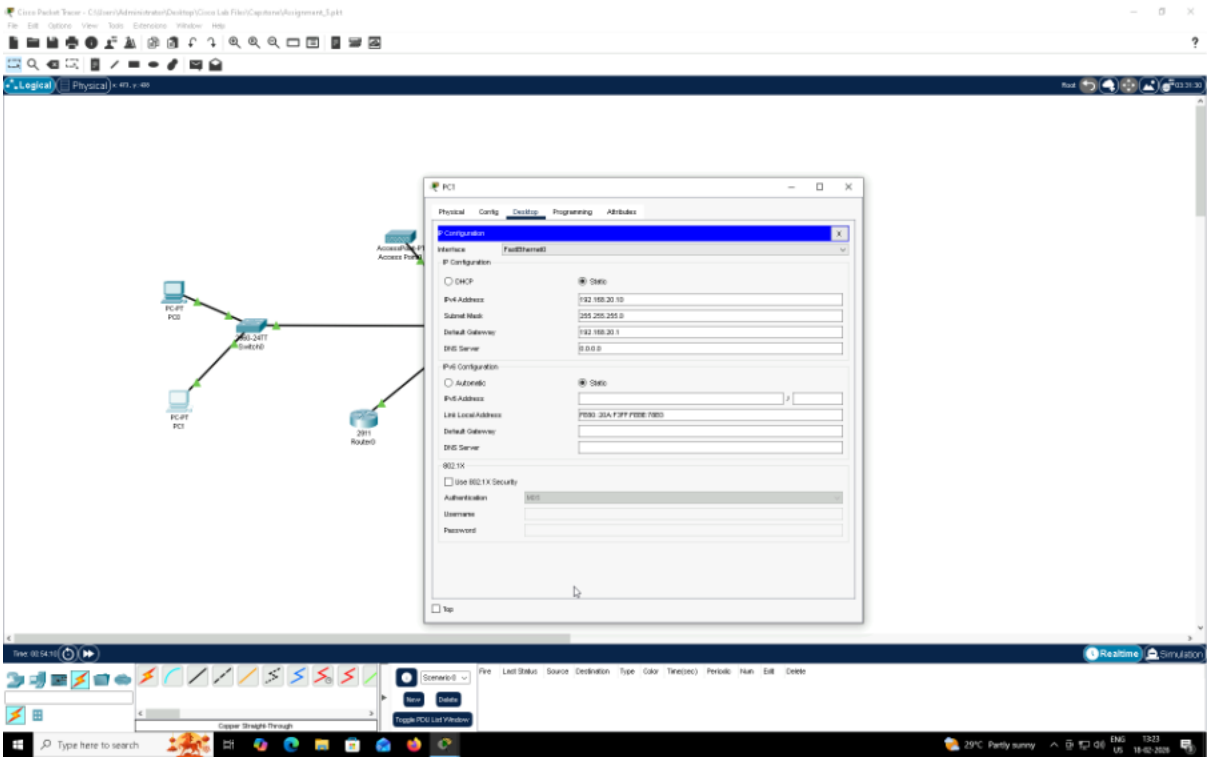
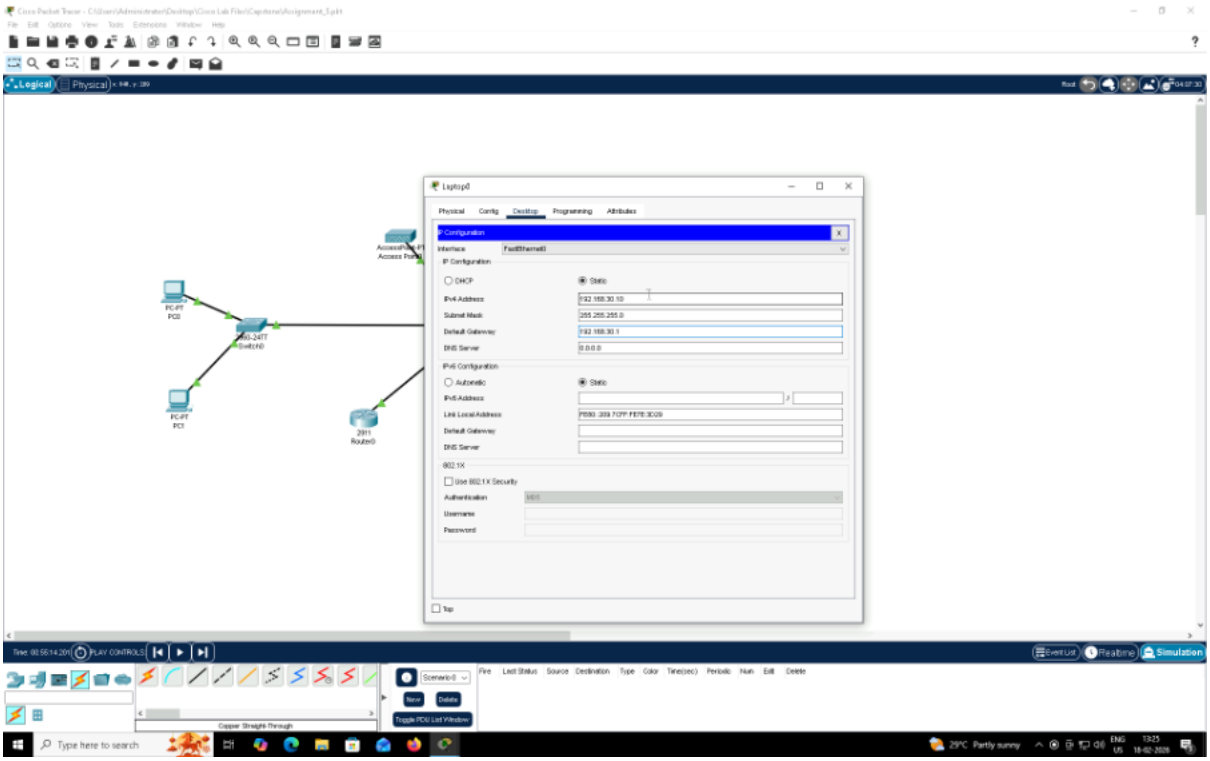
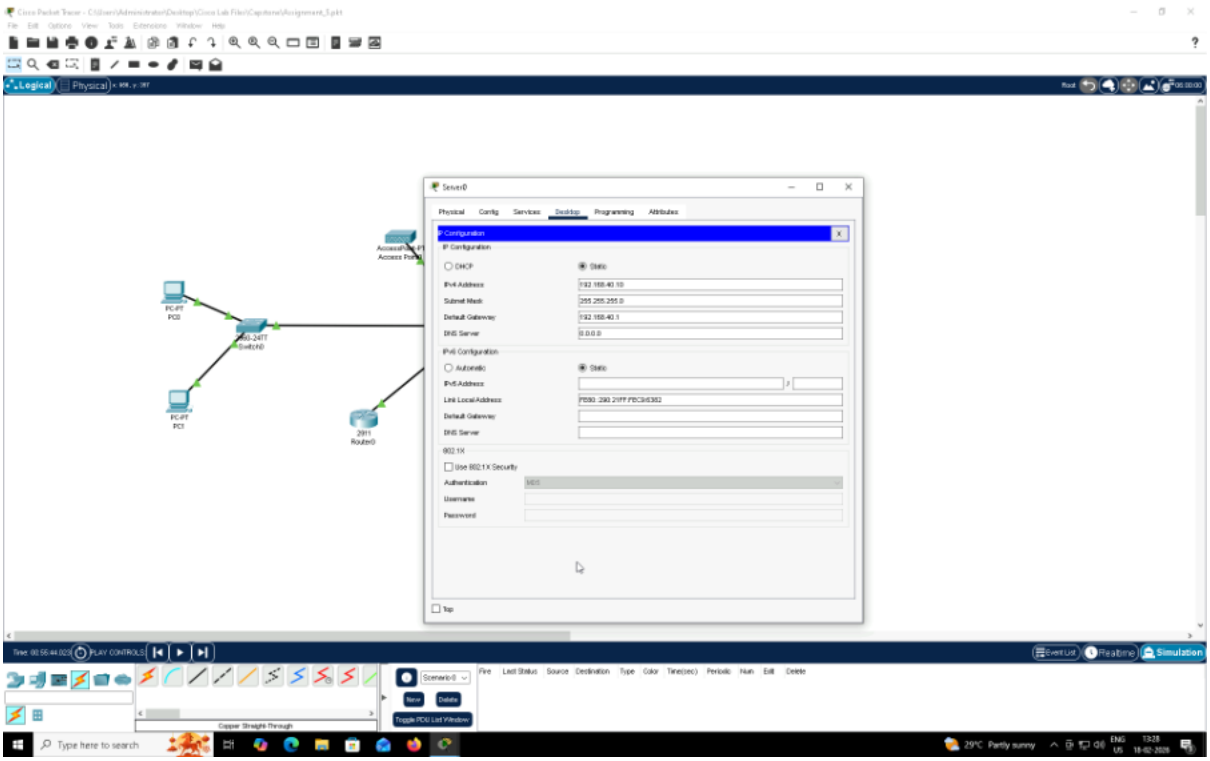
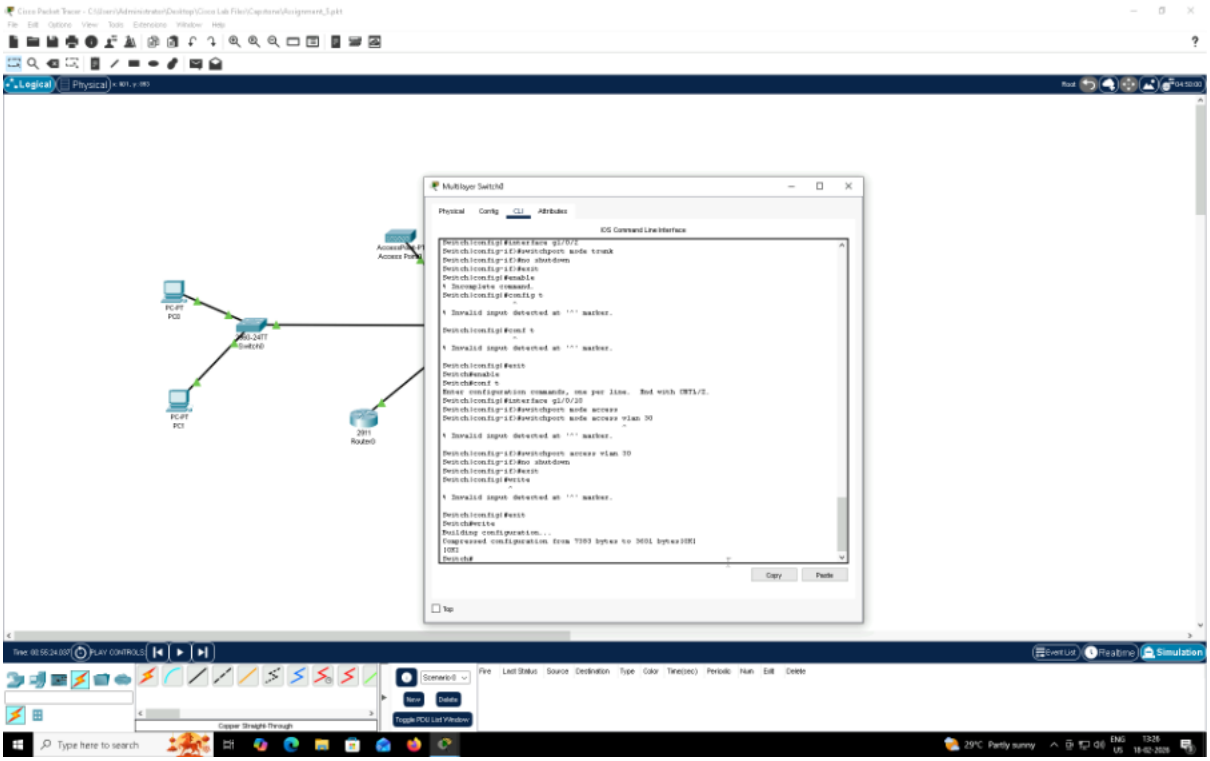


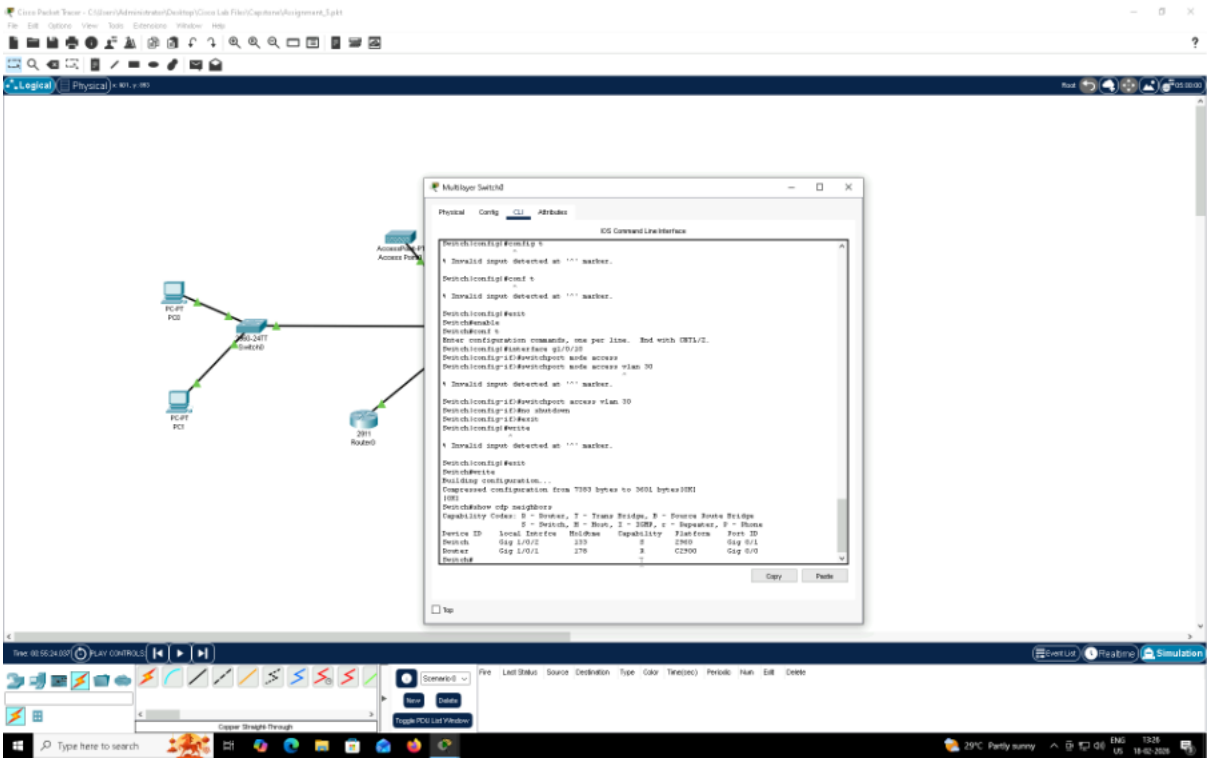


The wireless client was configured with an IP address in the VLAN 30 subnet and the default gateway set to the SVI address of VLAN 30.

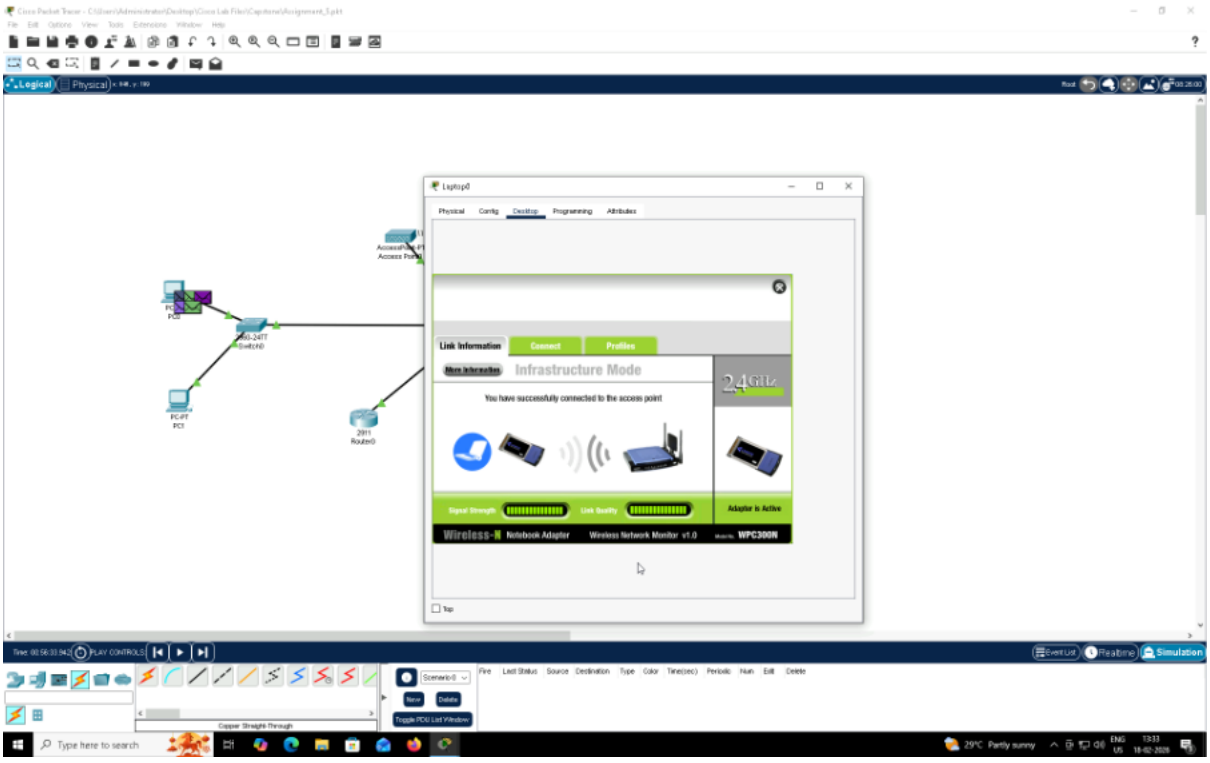


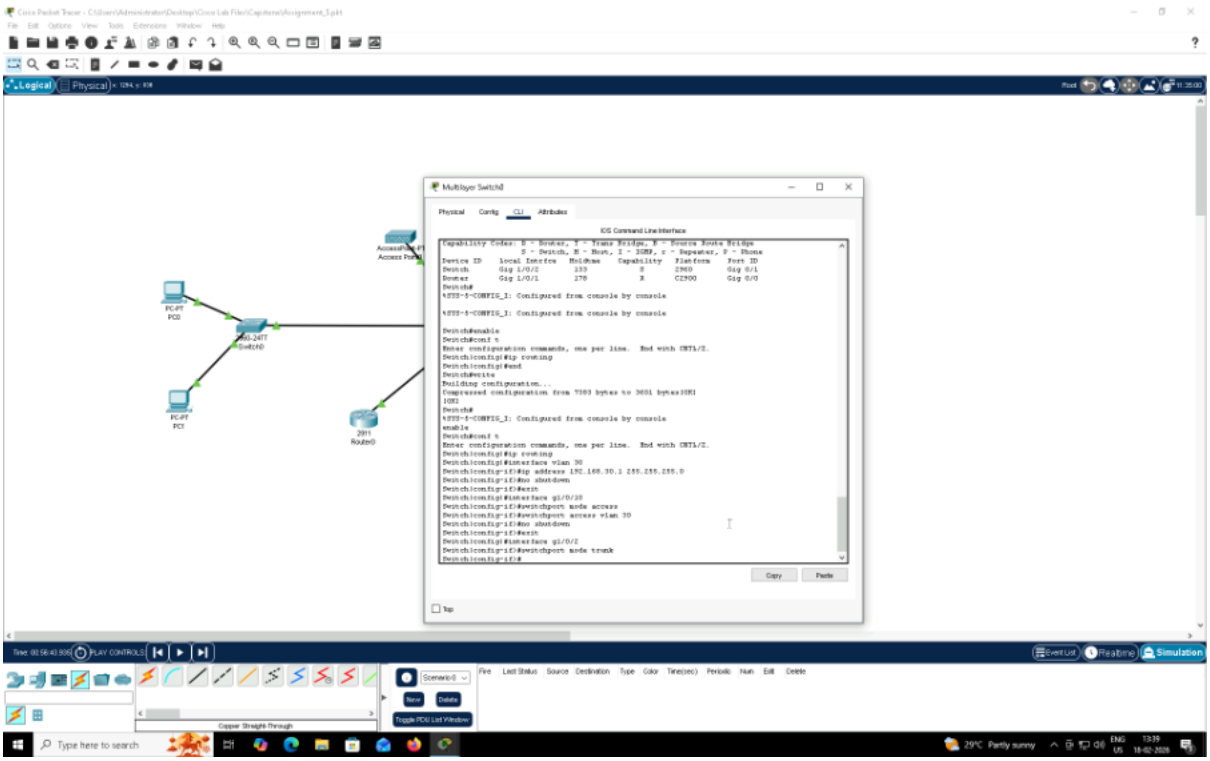
Static IP addressing was configured on end devices. The default gateway of each device was set to the corresponding VLAN interface IP on the multilayer switch, enabling layer-3 routing between VLANs.

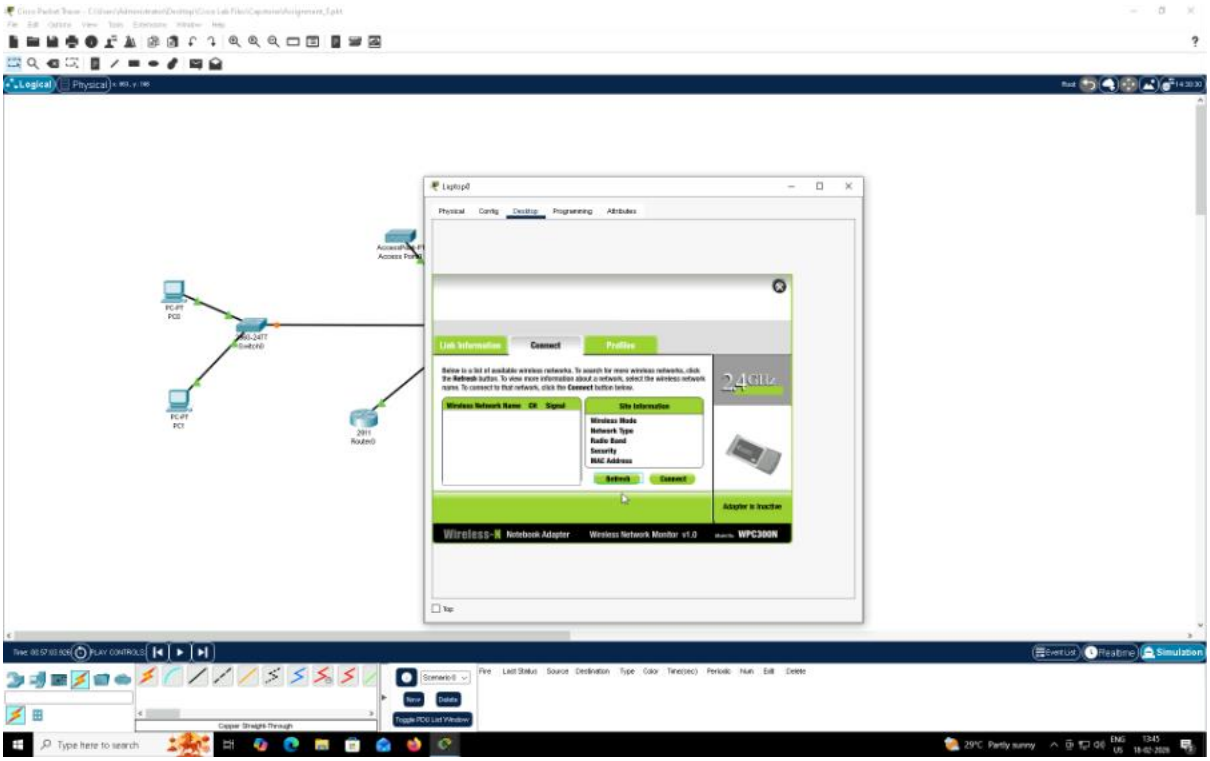


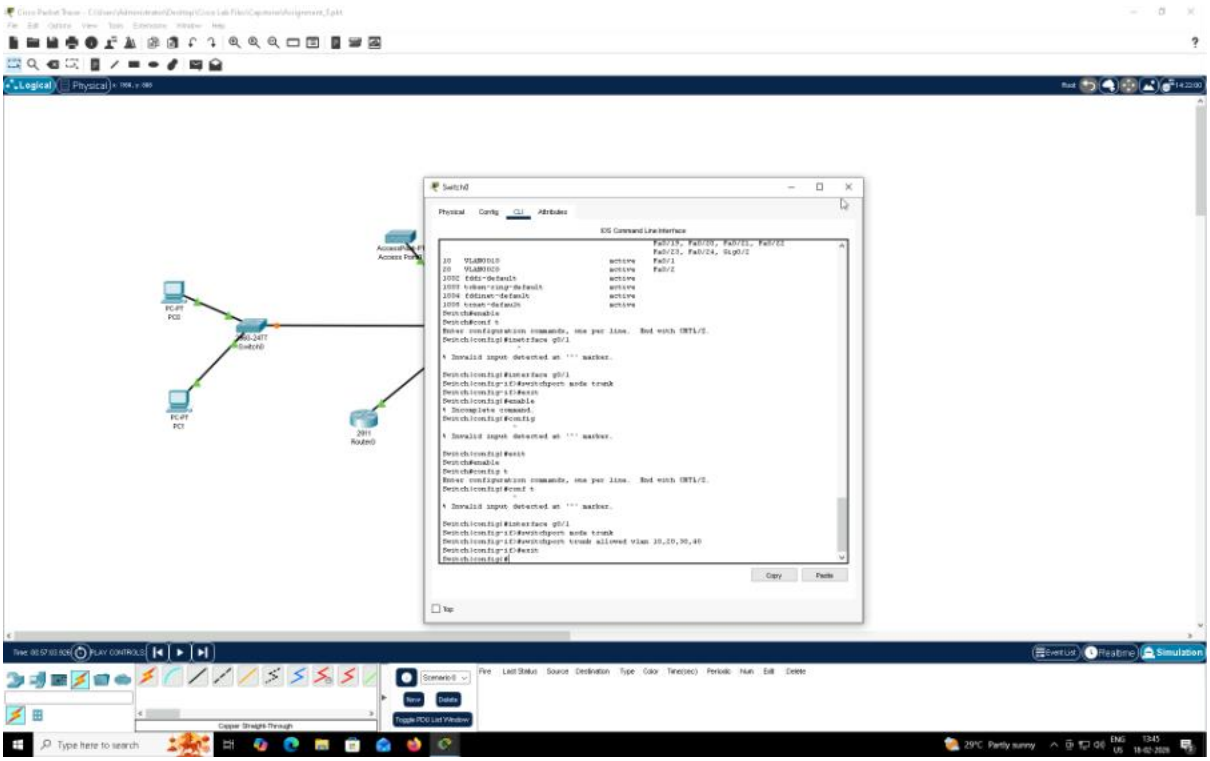


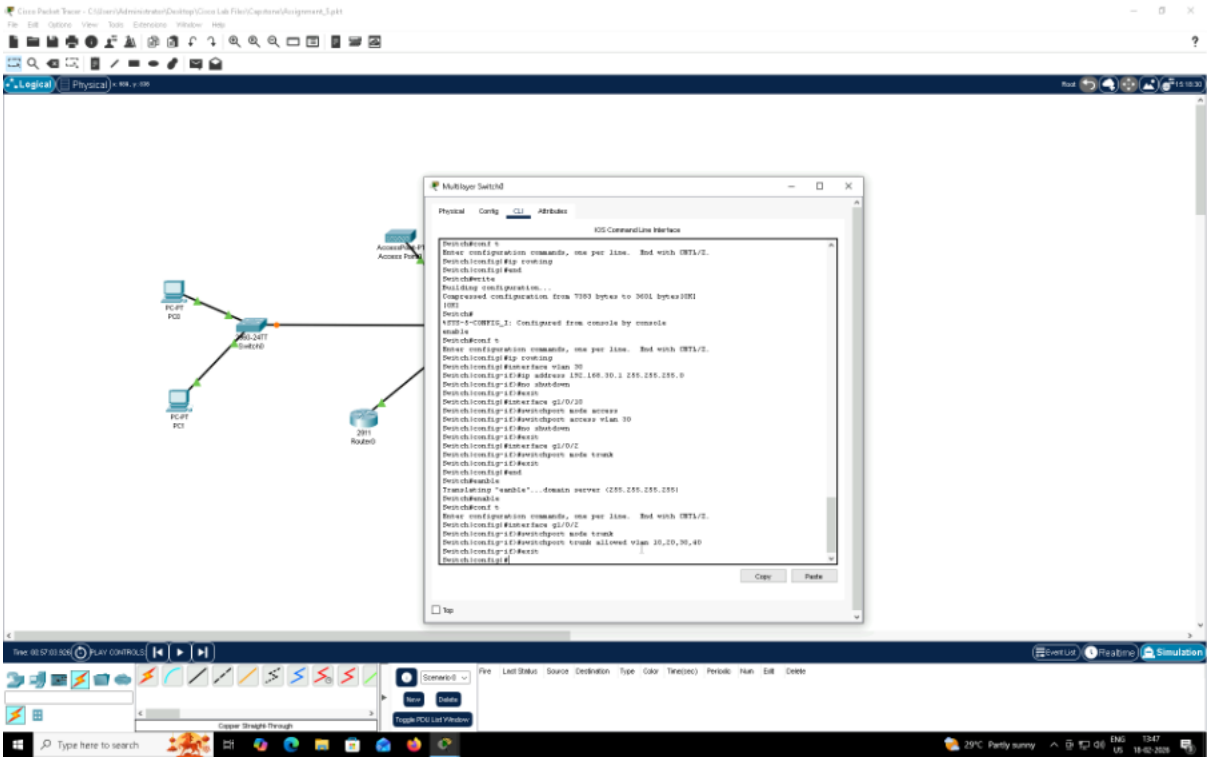
The wireless access point was assigned to VLAN 30 by configuring the connected switch interface as an access port in VALN 30. This separates wireless users from wired networks.

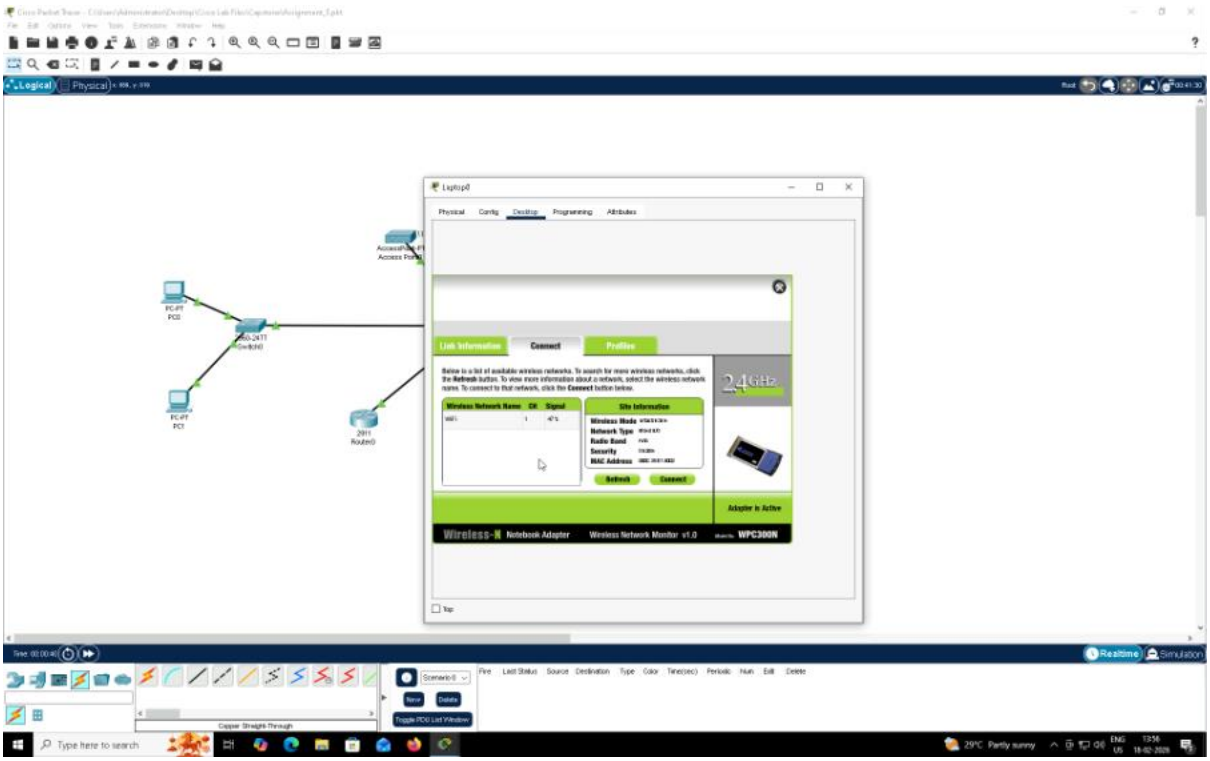


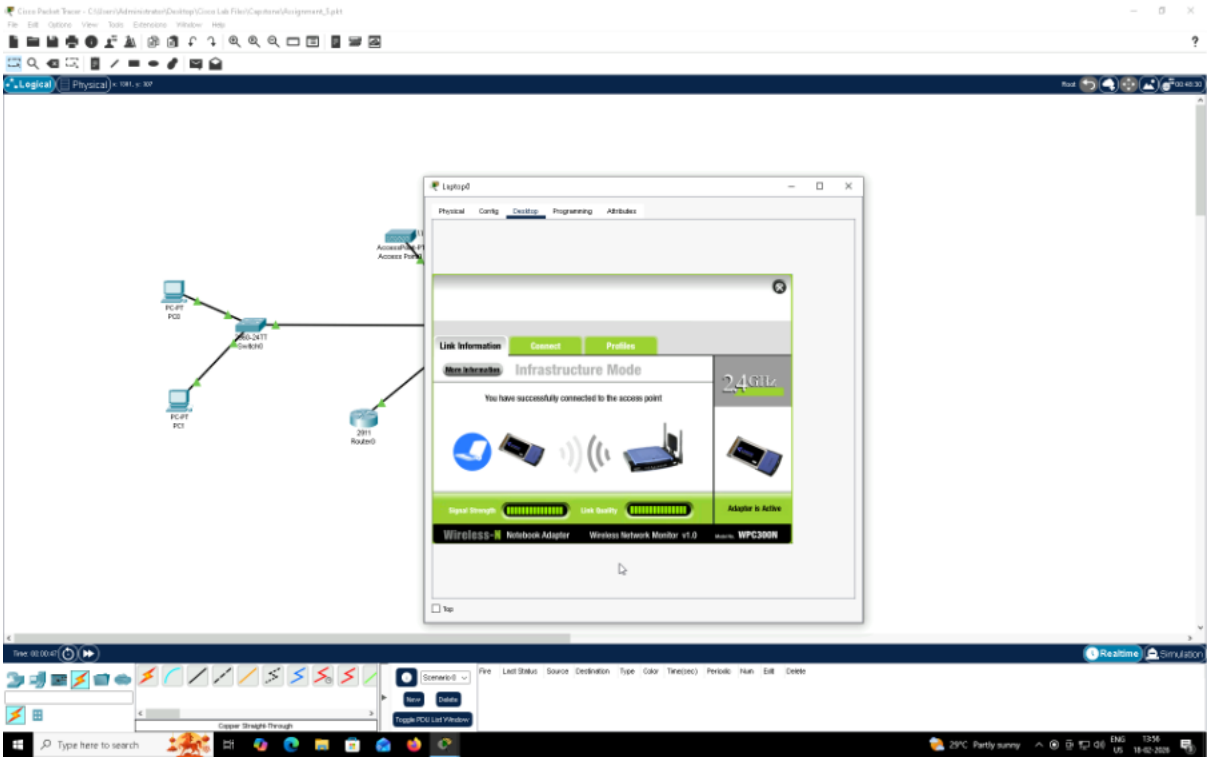

Multilayer Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Copy
Paste
Top
PC-PT PC0
PC-PT PC1
2960-24TT
2911 Router0
Logical
Physical
Simulation
Type here to search

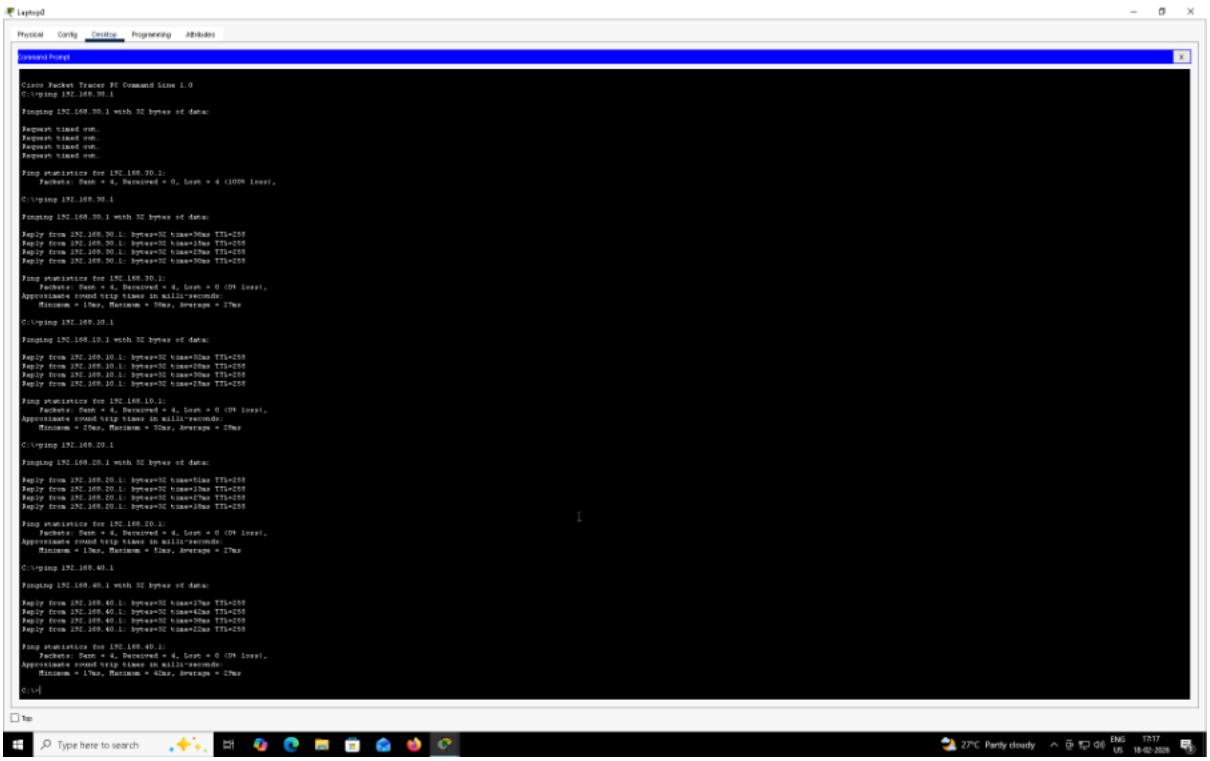

Laptop0
Physical
Config
Desktop
Programming
Attributes
Link Information
Connect
Profiles
24GHz
Site Information
Wireless Mode
Network Type
Radio Band
Security
MAC Address
Refresh
Connect
Adapter is Inactive
Wireless-N Notebook Adapter
Wireless Network Monitor v1.0
WPC300N
PC-PT PC0
PC-PT PC1
2960-24TT
2911 Router0
Logical
Physical
Simulation
Type here to search



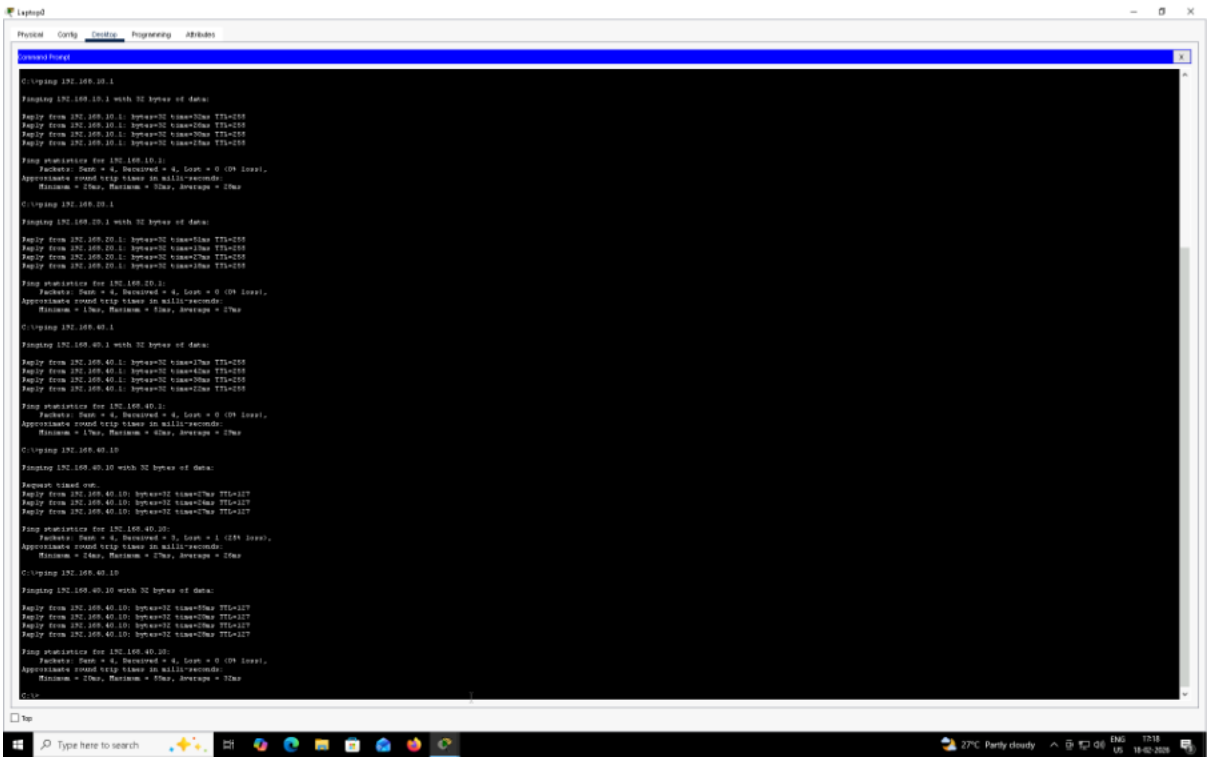



Laptop0
Physical
Config
Desktop
Programming
Attributes
Link Information
Connect
Profiles
Site Information
Refresh
Connect
2.4GHz
Adapter is Active
Wireless-N Notebook Adapter
Wireless Network Monitor v1.0
WPC300N
Logical
Physical
Realtime
Simulation
Copper Straight-Through
Type here to search


Laptop0
Physical
Config
Desktop
Programming
Attributes
Link Information
Connect
Profiles
Infrastructure Mode
You have successfully connected to the access point
2.4GHz
Adapter is Active
Wireless-N Notebook Adapter
Wireless Network Monitor v1.0
WPC300N
Logical
Physical
Realtime
Simulation
Copper Straight-Through
Type here to search



Connectivity tests were performed between devices located in different VLANs. Successful ICMP replies confirmed that the multilayer switch performed routing between VLSN networks correctly.



The wireless client connected through the access point successfully communicated with the wired hosts and the server, confirming proper VLAN assignment and trunk configuration.

Conclusion: -

Through the networking assignments, practical knowledge of network configuration and troubleshooting was gained. Concepts such as IP addressing, VLAN configuration, inter-VLAN routing using Layer 3 switches, router configuration, static routing, and network connectivity verification using ping commands were implemented.

The exercises improved understanding of how devices communicate within and across subnets, and how routing enables connectivity between different network segments.

Overall, the assignments strengthened foundational networking skills and enhanced the ability to design, configure, and troubleshoot enterprise network environments.